

Cryptography - 19_12

② 1

Let f be a length-preserving OWF, and h be hard-core for f . Establish if the following is a secure PRG:

$$G(x) = f(x) || h(x)$$

Recall: We have proven that indeed G is a PRG if f is a OWF.

The proof: We consider $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$

$$G(U_n) \equiv f(U_n) || h(U_n) \approx_c f(U_n) || U_1 \equiv U_n || U_1 \equiv U_{n+1}$$

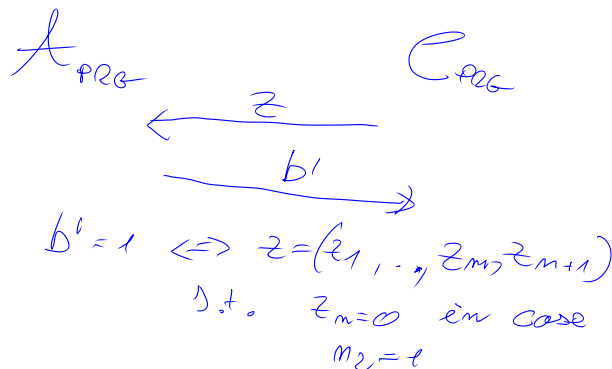
It doesn't work. To disprove it we must give $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ such that f is OWF but G is not secure.

For instance, we could take $f(x) = f(x_1, x_2)$ (with $|x_1| + |x_2| = n$), and $= g(x_1) || 0^{|x_2|}$ where $g : \{0, 1\}^{n_1} \rightarrow \{0, 1\}^{n_1}$ is OWF.

Now, g OWF $\implies f$ OWF.

$$G(x) = f(x) || h(x) = g(x_1) || 0^{n_2} || h(x)$$

is not a PRG.



② 2

Let $\mathbb{G}$ be a cyclic group, with generator g and prime order q . Assume it's possible to compute $\sqrt{\cdot}$ over $\mathbb{G}$ if it exists.

Prove that CDH is equivalent to:

SQUARE-DH: Given (g, g^a) for $a \in \mathbb{Z}_q$ output g^{a^2}

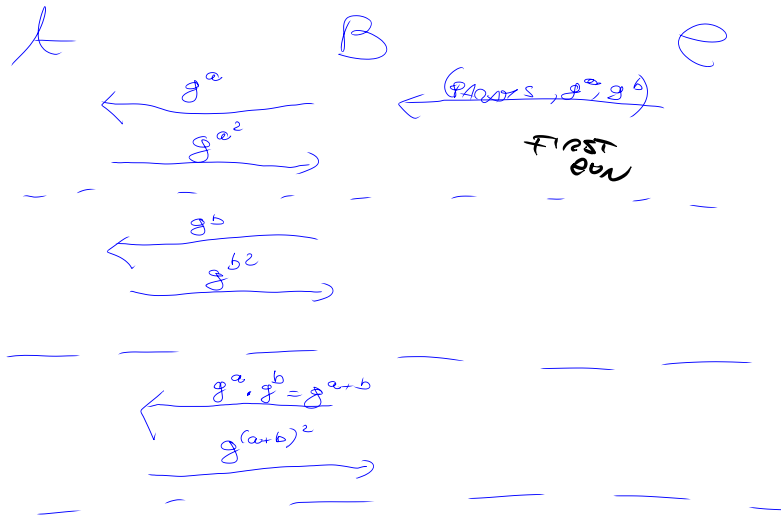
We must prove:

1. CDH $\implies$ SQUARE-DH

2. SQUARE-DH $\implies$ CDH

For the first, $\exists$ PPT $\mathcal{A}$ such that $\Pr[\mathcal{A}(\text{params}, g^a) = g^{a^2}] \geq \frac{1}{\text{poly}}$

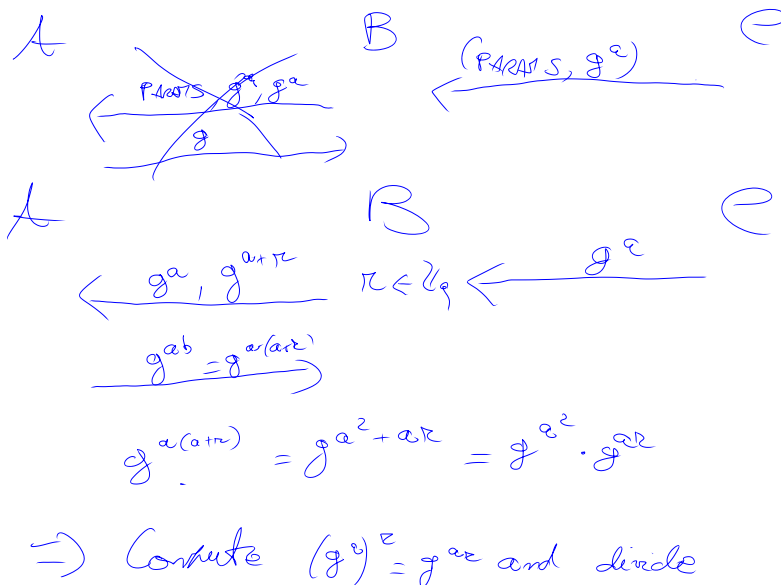
Then build PPT $\mathcal{B}$ against CDH.



$\implies$ with probability $\left(\frac{1}{\text{poly}}\right)^3$ we know g^{a^2} , g^{b^2} , and $g^{(a+b)^2}$

Then $\frac{g^{(a+b)^2}}{g^{a^2} \cdot g^{b^2}} = g^{2ab} = (g^{ab})^2$

Now, let's try part two.



Let $\Pi = (KGen, Enc, Dec)$ be a PKE scheme for $\mathcal{M} = \{0, 1\}$. Build Π' a PKE scheme for $\mathcal{M} = \{0, 1\}^l$ for $l(\lambda) = \text{poly}(\lambda)$.

Assume Π is CPA-secure. Π' also should be CPA-secure.

Suggestion: $(pk, sk) \leftarrow KGen(1^\lambda)$ and given $m = m[1], m[2], \dots, m[l]$ output $c = c_1, c_2, \dots, c_l$ and $c_i = Enc(pk, m_i)$.

Proof of CPA-security. Assume not, $\exists$ PPT $\mathcal{A}$ that breaks CPA security.

We need hybrid arguments.

Let $\square \equiv Enc(pk, \cdot)$.

Challenge:

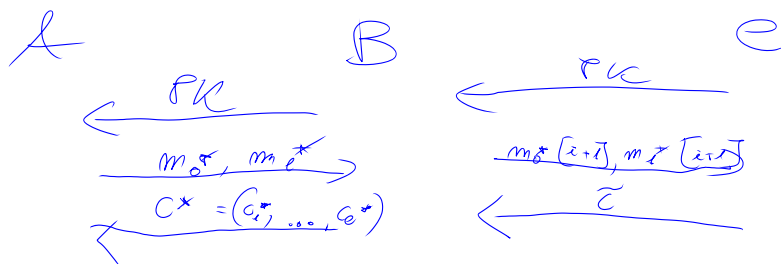
$$H_0 \equiv \boxed{m_1[1]^*}, \dots, \boxed{m_1^*[l]}$$

$\vdots$

$$H_i \equiv \boxed{m_0^*[1]}, \dots, \boxed{m_0^*[i]}, \boxed{m_1^*[i+1]}, \dots, \square$$

$\vdots$

H_l with all m_0 's.



$c_1^*, \dots, c_i^*$ all $Enc(pk, m_0^*[j])$

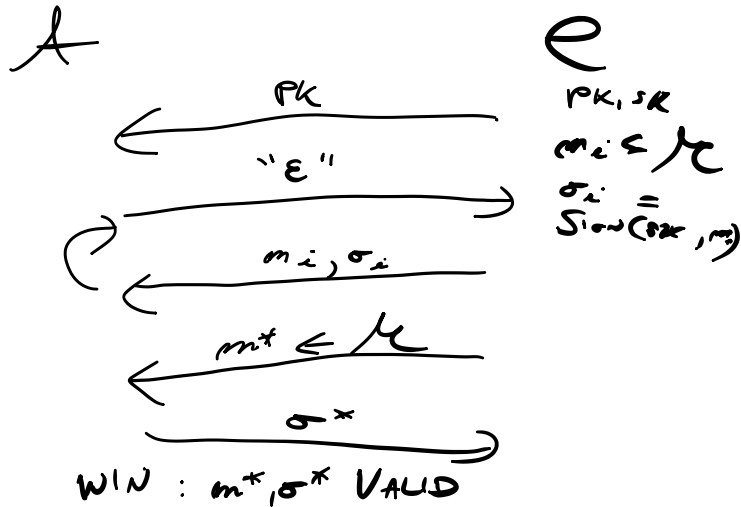
$c_{i+1}^* = \tilde{c}$

$c_{i+2}^*, \dots, c_l^*$ all $Enc(pk, m_1^*[j])$

④

Consider RUF-RMA:

$\text{GAT} \in \text{RUF-RMA}_{\Pi, \mathcal{L}}(1)$



$$|\mathcal{M}| = u(\log l)$$

Question: $\text{UF-CMA} \implies \text{RUF-RMA}$? And $\text{RUF-RMA} \implies \text{UF-CMA}$.

The first is trivial. Work out the reduction.

Two is false. Take any UF-CMA Π and consider Π' with $|\mathcal{M}'| = 2|\mathcal{M}|$.

$\sigma' = \text{Sign}(sk, m')$.

Say $m' = m_1 || m_2$ with $|m_1| = |m_2| = n$.

Check: if $m_1 = 0^n$, $\sigma' = m_2$. Else $\sigma' = \text{Sign}(sk, m_2)$.

Alternative: Let $pk' = (pk, \bar{m}, \bar{\sigma})$ where $\bar{m} \in \mathcal{M}$, $\bar{\sigma} = \text{Sign}(sk, \bar{m})$.

5

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a OWF.

Consider the following $\Pi = (\text{KGen}, \text{Sign}, \text{Vrfy})$.

$\text{KGen}(1^\lambda) : (x_i^0, x_i^1)_{i \in [l]}$ with $x_i^0, x_i^1 \leftarrow \{0, 1\}^n$.

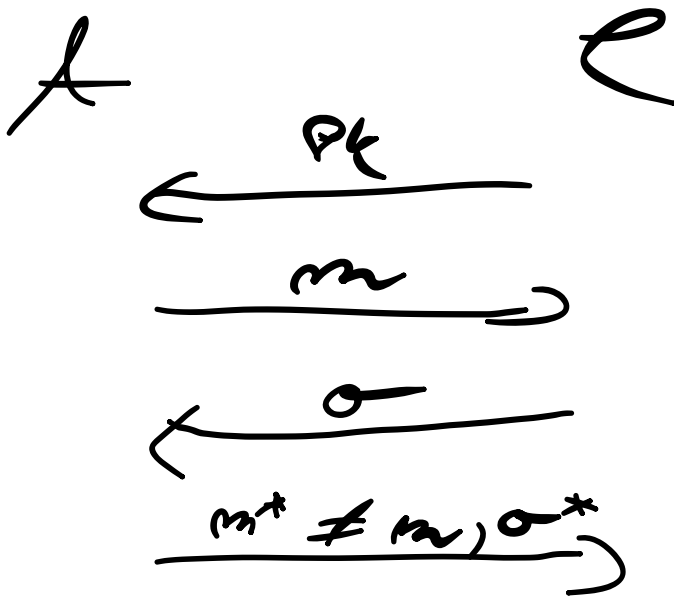
$pk = (y_i^0, y_i^1)_{i \in [l]}$

$\forall b \in \{0, 1\}, y_i^b = f(x_i^b)$

$\text{Sign}(sk, m = m[1], \dots, m[l])$. Output $\sigma = (x_i^{m[i]})_{i \in [l]}$.

Vrfy : Just check that $f(x_i)^{m[i]} = y_i^{m[i]} \forall i$

Prove Π is one-time secure assuming f is OWF.



Observation: $\exists j$ such that $m^*[j] \neq m[j]$.

For this position j , σ contains x_j^0 a pre-image of y_j^0 but σ^* contains x_j^1 , a pre-image of y_j^1 . This suggests the following reduction:

